7	**(a)**	State experimental evidence to suggest that the process of radioactive decay is		
		(i)	random;	
			……………………………………………………………………………………… ………………………………………………………………………………………	[1]
			The points on a <u>count rate against time graph</u> <u>scatter about the (exponential) line of best fit</u>. **OR** **<u>Measured count rate fluctuates</u>** from instant to instant in time**.**	
		(ii)	spontaneous.	
			……………………………………………………………………………………… ………………………………………………………………………………………	[1]
			The <u>rate of decrease of the measured count rate</u> is unaffected by external stimuli and changes in physical conditions. **OR** **<u>Repeated experiments</u> under different physical conditions result in no change in the <u>rate of decrease of the measured count rate</u>.** ***Answers must refer to an <u>observable measurement e.g. "measured count rate"</u>.***	
	(b)	A student measured the count rate from a sample of pure iodine-131 at various times. The results are shown in Fig. 7.1. Iodine-131 decays by beta emission to form xenon-131 which is stable.		

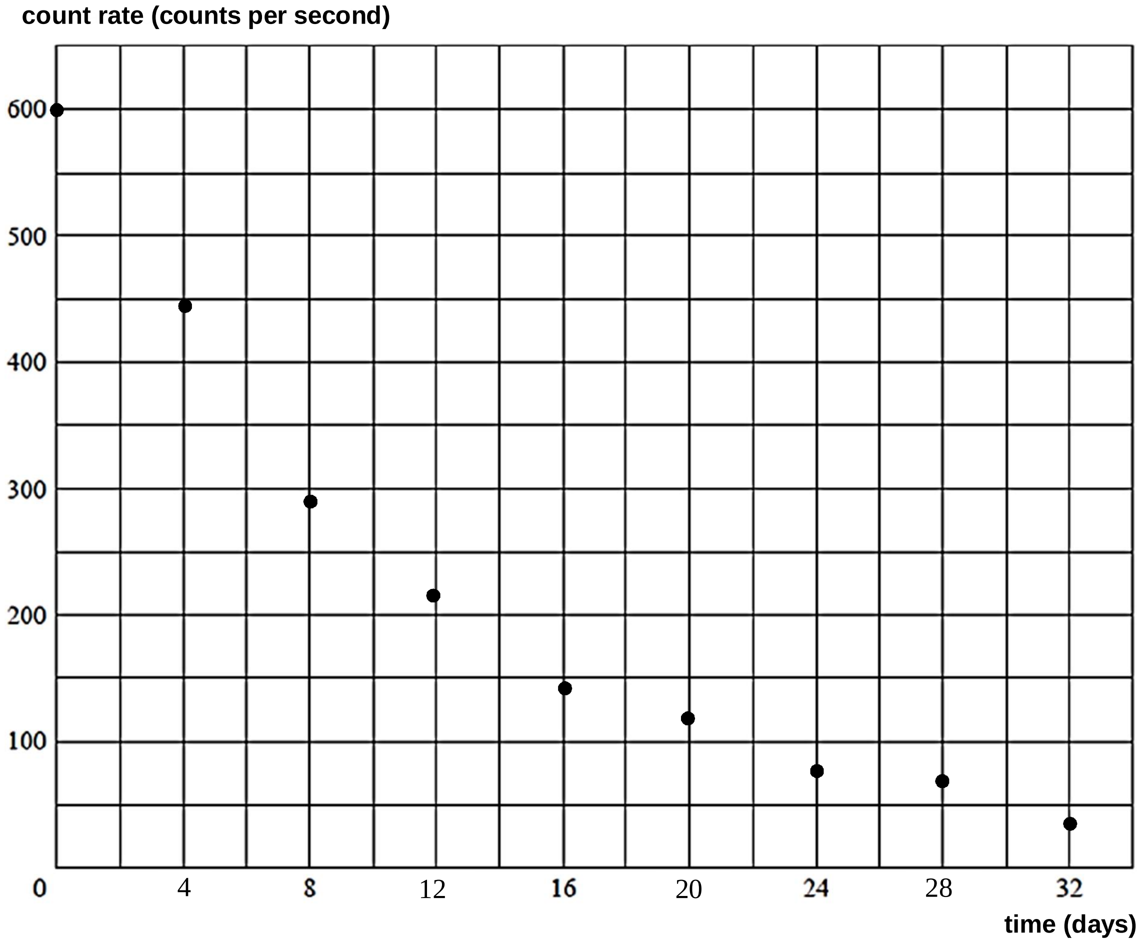
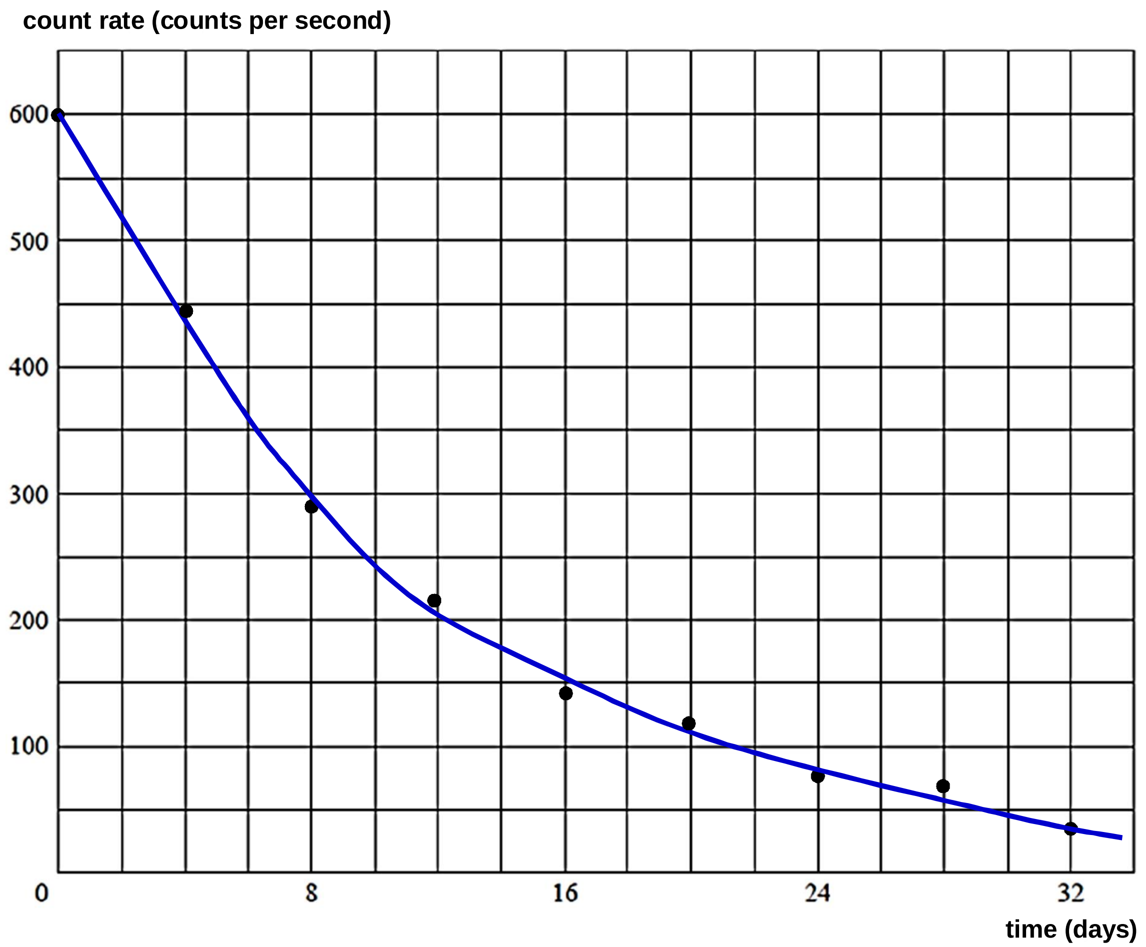


Fig. 7.1

		Use Fig. 7.1 to determine a value for the half-life, in days, of iodine-131.	

Solution:



- **Draw a curve of best-fit on Fig. 7.1.**

- **Use coordinates from best-fit curve to determine the average time taken for count rate to decrease to half the original value.**

e.g.
Read of time taken for count rate to decrease from 600 s^{-1} to 300 s^{-1} → $t_{1/2,\,1}$
Read of time taken for count rate to decrease from 300 s^{-1} to 150 s^{-1} → $t_{1/2,\,2}$
Calculate the average of $t_{1/2,\,1}$ and $t_{1/2,\,2}$.
[Working for determining the half-life from the average of at least 2 half-lives must be shown.]

- **Answer: Accept (8 days ± 1 day). [Actual half-life is about 8.06 days.]**

Alternative working:
Accept use of $C = C_o\, e^{-\lambda t}$ and $C = C_o\, (\tfrac{1}{2})^{t/t_{1/2}}$ provided coordinates are read off the best-fit curve and not the raw data points.
Also, two values of half-life should be computed from at least two different coordinates that are far apart on the best-fit curve, and average calculated.

half-life = ………………… days [3]

	(c)	Explain why the determination of the half-life of iodine-131 by the method in **(b)** requires that the product of the decay is stable.	
		……………………………………………………………………………………… ………………………………………………………………………………………	[1]
		If the product is unstable, it will decay into another product and emit radiation in the process, and so **the measured count rate would be higher than the actual count rate due to iodine-131 only.** **OR** **the measured count rate would not be due to the decay of iodine-131 only.**	
	(d)	Suggest why the percentage systematic error will not be significant if the student neglected the background radiation.	
		……………………………………………………………………………………… ………………………………………………………………………………………	[1]
		From the graph, the count rate is almost zero eventually; indicating that the background radiation is insignificant (or much smaller) compared to the count rates in this experiment (in the order of hundreds per second). **OR** **Background radiation is typically less than 1 count per second (or around 0.5 counts per second or 30 counts per minute), which is much smaller compared to the count rates in this experiment (in the order of hundreds per second).**	